

ADARSH KUMAR

Staff Software Engineer · Fintech & Payment Infrastructure

adarshkumar2k@gmail.com · +60 13 530 8971 · Kuala Lumpur, Malaysia · [LinkedIn](#) · [Portfolio](#)

Professional Summary

Senior software engineer with 10 years of experience building and scaling regulated fintech platforms, payment systems, and crypto exchanges across Southeast Asia. Specialized in distributed systems, payment infrastructure, and high-throughput backend services. Track record of taking systems from zero to millions of users, co-founding a venture-backed startup, and leading engineering teams across multiple countries. Strong IC with deep technical acumen and a product-oriented mindset.

Work Experience

StashAway · Kuala Lumpur, Malaysia

Senior Staff Engineer (Apr 2024 – Present)

- **Platform modernization:** Decomposed monolith serving 500K+ users across 5 SEA markets into 12+ microservices with Kafka event messaging; zero-downtime migration cut release cycles 40% and enabled 3x traffic spike handling during market volatility.
 - **Money movement & AML compliance:** Own end-to-end deposit, cash-in, and withdrawal flows for 500K+ users across 5 SEA markets; built AML transaction monitoring, screening, and regulatory reporting aligned with each market's central bank framework.
 - **Onboarding conversion:** Consolidated KYC/auth from 8-screen to 4-screen flow with OAuth2 + MFA and parallelized background checks; signup time down 35%, conversions exceeded target by 10%.
 - **AI-assisted dev tooling:** Introduced MCP-based automated code review and test generation, clearing senior-engineer PR review bottleneck; review turnaround down 50%, team shipping speed up 25%.
 - **API design & platform contracts:** Defined and enforced API standards across 12+ services (versioned contracts, OpenAPI specs, consistent error handling, pagination); eliminated cross-team integration friction.
 - **Infrastructure cost reduction:** Cut AWS staging costs 30% (~\$180K/year) via automated resource scheduling, right-sizing with Datadog utilization metrics, and Terraform cost tagging.
 - **Disaster recovery:** DR playbook covering database failover, cross-region backup, and graceful degradation; established on-call rotation and structured incident review, cutting MTTR.
 - **Mentorship & team growth:** Built a structured program for 8 engineers: weekly architecture walkthroughs, pair programming rotations, and monthly tech talks.
 - **Cross-functional collaboration:** Worked with product, compliance, and data science across 5 countries; translated regulatory requirements in different SEA markets into platform features without slowing delivery.
- Stack:** Node.js, React Native, Go, Temporal, Kafka, PostgreSQL, MongoDB, Redis, AWS, Kubernetes, Datadog, Sentry

Xendit (YC S15) · Philippines, Indonesia & Malaysia

Technical Lead (Sep 2022 – Feb 2024) · Senior Software Engineer (Feb 2021 – Aug 2021)

- **Payment routing engine:** Built a real-time routing engine with channel health scoring and automatic failover across 30+ bank/e-wallet integrations in 3 countries on \$15B+/year payment volume; processing time 8s → 2.4s, success rate 88% → 97%.
- **Circuit breakers & resilience:** Built fault isolation layer with channel circuit breakers and graceful degradation during 10K+ QPS peaks; failed transactions down 250%, platform availability 99.99%.
- **Storage & cost optimization:** Optimized 500M+ transaction records with S3 lifecycle tiering, DB indexes, and analytics read replicas; S3 costs down 65%, DB costs down 40%, p99 latency under 200ms.
- **Unified notification service:** Built transactional service for WhatsApp, SMS, Viber, and email with automatic fallback chain; 99.5% delivery rate across 10M+ monthly notifications.
- **Reconciliation & idempotency:** Implemented payout retry and reconciliation using outbox pattern and idempotency keys (PCI-DSS aligned); zero lost or double-processed transactions even during partner bank outages.
- **Observability:** Built real-time Datadog/Grafana dashboards and alerting for payment success rates, partner health, and circuit breaker state across all channels.
- **Team leadership:** Led 6-person engineering team; introduced TDD, code review standards, and sprint retros; delivery velocity up 35% over two quarters. Led technical hiring.

- **Architecture Review Group:** Core member; reviewed cross-team system designs and drove alignment on patterns; cut cross-team review time 80% via structured templates and async feedback.

Stack: Node.js, Go, PostgreSQL, MongoDB, Redis, Kafka, AWS, Datadog, Sentry, Splunk, Databricks, Redash

On Demand Deals · Manila, Philippines

Co-Founder & CTO (July 2021 – Sep 2022)

- **Founder to product-market fit:** Co-founded a SaaS platform for F&B businesses; owned all technical decisions, hiring, and product delivery end-to-end.
- **Full-stack platform build:** Built entire platform solo initially, then hired and managed a team of 4 engineers; kept stack deliberately simple to move fast.
- **Fundraising:** Closed \$480K pre-seed from Iterative (YC for Southeast Asia) and Accelerating Asia; led financial model and technical DD.
- **Growth:** Scaled to 50+ shop locations across Manila in 6 months; \$1.37M GMV with 200% MoM growth.
- **Partner integrations:** Managed relationships with payment providers, POS integrators, and delivery partners; negotiated technical integrations and commercial terms.

Stack: React, React Native, Node.js, PostgreSQL, AWS, Terraform, Kafka

PDAX (Philippines Digital Assets Exchange) · BGC, Philippines

Senior Software Engineer (Nov 2019 – Feb 2021) · Software Engineer (Mar 2018 – Nov 2019)

- **Founding engineer:** Joined pre-platform; core team that grew PDAX into a top-3 BSP-regulated crypto exchange in the Philippines.
- **Trade matching engine:** Built a core engine handling 5K+ concurrent orders at sub-100ms matching latency; designed for horizontal scaling during crypto market volatility.
- **Bank & e-wallet integrations:** Connected major Philippine banks (BDO, BPI, UnionBank) and e-wallets (GCash, PayMaya) for fiat deposits/withdrawals; settlement 3 days → real-time.
- **LOCQ product launch:** Designed and launched first-of-its-kind blockchain-based fuel futures product (prepaid gas at today's prices on-chain); drove a wave of non-crypto-native users.
- **Wallet & ledger system:** Built internal wallet and ledger infrastructure with double-entry accounting across crypto and fiat; full auditability for BSP regulatory compliance and KYC/AML monitoring.
- **Risk & compliance tooling:** Built an admin dashboard and real-time monitoring for suspicious trading patterns, large withdrawals, and KYC checks.
- **Reliability under load:** Optimized trade engine and API layer with connection pooling, Redis query caching, and graceful degradation; 99.9%+ uptime during major market events.

Stack: Node.js, PostgreSQL, TimescaleDB, Redis, React Native, Solidity, AWS, Web3, ERC20

Primus Software Corporation · Greater Noida, India

Software Engineer (Sep 2017 – Mar 2018)

- **Enterprise applications:** Built web/mobile apps for healthcare, logistics, and education clients (React, Angular, Node.js, PostgreSQL); led 3 pre-sales PoCs that won client contracts.

Technical Skills

Languages: JavaScript, Typescript, Python, Go, Solidity

Backend/Frontend: Node.js, NestJS, Temporal Workflows, REST, gRPC, n8n, React, React Native, Redux

AI & LLM: MCP, RAG pipelines, vector DBs (Pinecone, pgvector), LLM context management, agentic workflows

Data & Storage: PostgreSQL, MongoDB, DynamoDB, Redis, Pinecone, TimescaleDB

Cloud & Infrastructure: AWS, Kubernetes, Terraform, Helm, Docker, CI/CD

Messaging & Streaming: Apache Kafka, RabbitMQ, AWS SQS, Flink

Observability: Datadog, Grafana, PagerDuty, OpsGenie, Sentry, Splunk

Data & Analytics: Databricks, Redash, Looker, Metabase

Education

Bachelor of Technology, Computer Science 2014 – 2018

G.L. Bajaj Institute of Technology & Management — Dr. A.P.J. Abdul Kalam Technical University, India